

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Y-Axis Glass Scale Replacement Procedure	Effective Date	25 Jan 2017

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	25 Jan 2017	First Release	TAN LIANG KAI

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A. Y-Axis Glass Scale

I. Y-Axis glass scale removal

S2EX machine Y-Axis glass scale assembly Part ID: 4XASP-A145

1. Turn off x-ray source and open front access door.
2. Remove spring clamps on encoder cover plate as shown in Figure 1.



Figure 1 Y-Axis glass scale

3. Unscrew screws as shown in Figure 2. Gentle remove y-axis cover plate and encoder cover plate from machine.

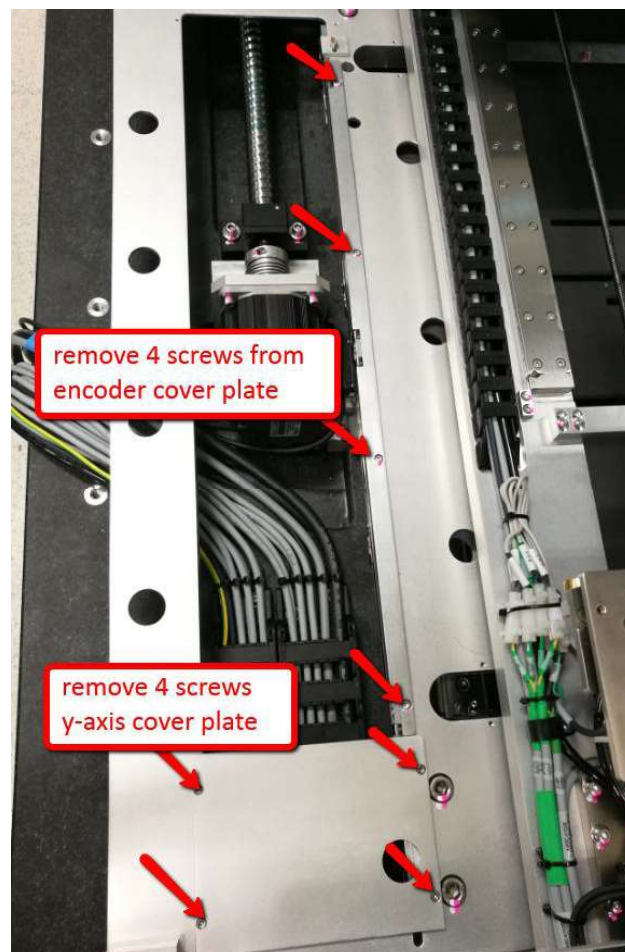


Figure 2 Screws removal

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4. Remove spring clamps, home sensor and limit sensor as shown in Figure 3.

Remark: Perform minor marking on current limit sensor & home sensor for ease of adjustment

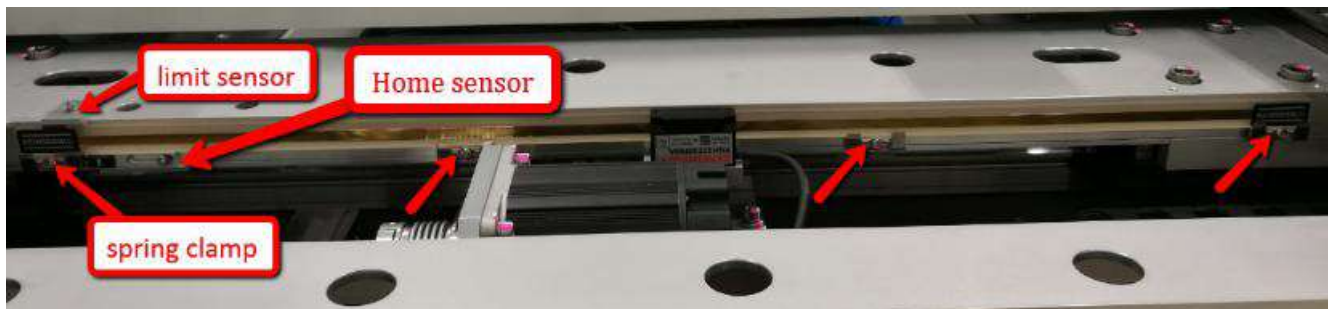


Figure 3 Glass scale

5. Gently remove Y-Axis encoder as shown in Figure 4 and allocate at visible position.



Figure 4 Y-Axis Encoder

6. Cut RTV glue which secure glass scale.

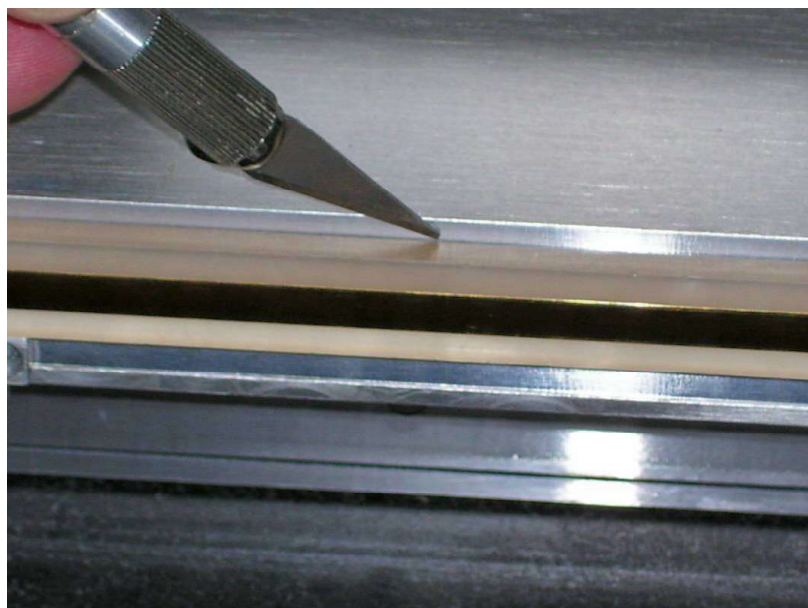


Figure 5 RTV glue removal

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- Gentle slide out Y-Axis glass scale toward up direction as shown in Figure 6.



Figure 6 Glass scale removal

II. Y-Axis Glass Scale Replacement

- Gentle assemble new Y-Axis glass scale as shown in Figure 7.

Remark: Caution on glass scale orientation.

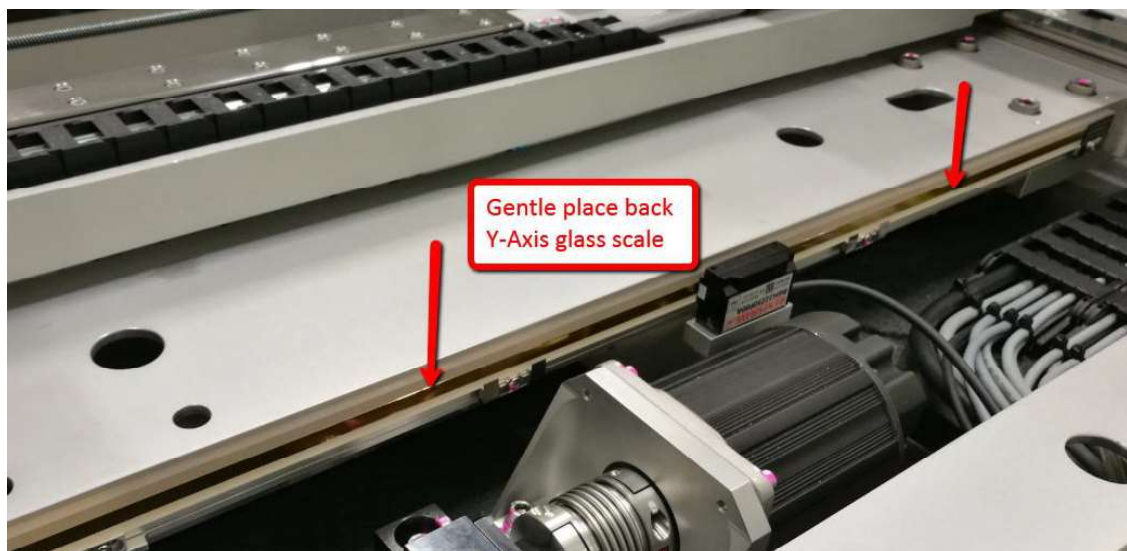


Figure 7 Glass scale replacement

- Position glass scale lining up at termination tag of the glass scale with spring clamp.



Figure 8 Glass scale lining up (Reference only)

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3. Assemble limit sensor and home sensor by refer to origin location (By refer to Step 4). Tighten all screws which secure spring clamp as shown in Figure 9.

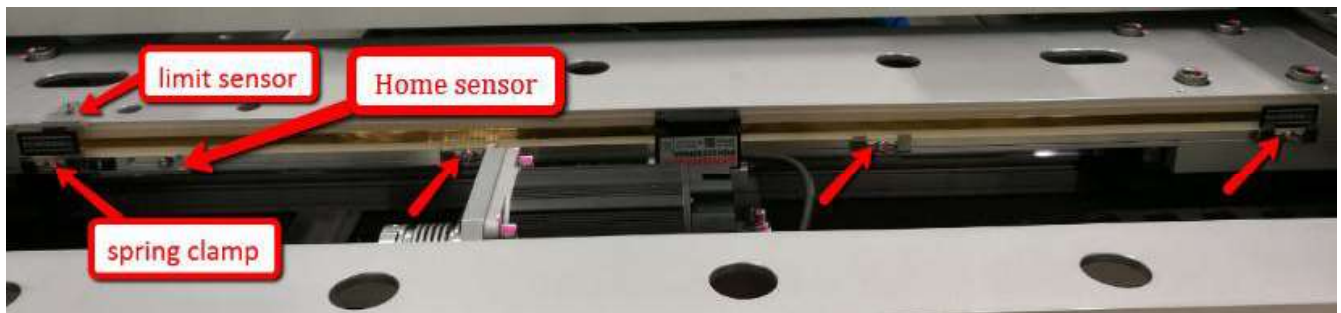


Figure 9 Glass scale replacement

4. Assemble Y-Axis encoder as shown in Figure 10.



Figure 10 Y-Axis encoder

5. Assemble Y-Axis cover plate and encoder cover plate as shown in Figure 11.

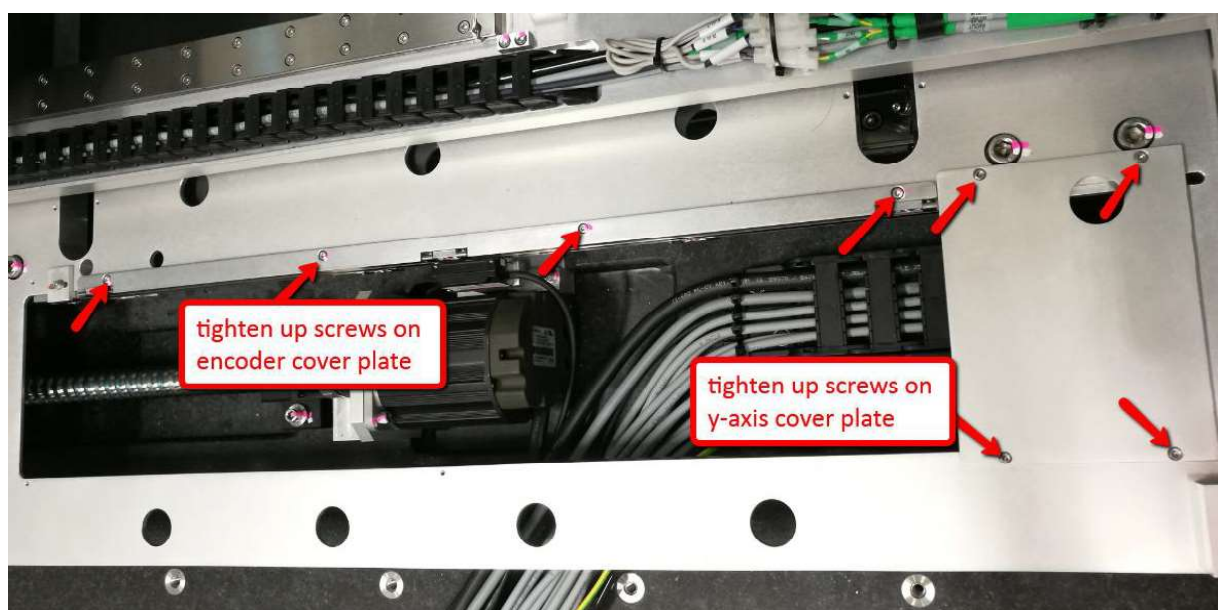


Figure 11 Cover plate assembly

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6. Tighten all screws which secure spring clamps on encoder cover plate as shown in Figure 12.

Remark: Make sure all spring clamps are well-positioned and do not contain any friction with encoder while moving the Y-Axis stage.



Figure 12 Cover plate assembly

7. Move Y-Axis stage to minimum hard stop (front side) and maximum hard stop (rear side). Visual check Y-Axis encoder which require green light illuminated along the way as shown in Figure 13. If adjustment require, May refer to “AXI V810 Y-Axis Read Head Encoder Replacement Procedure”.

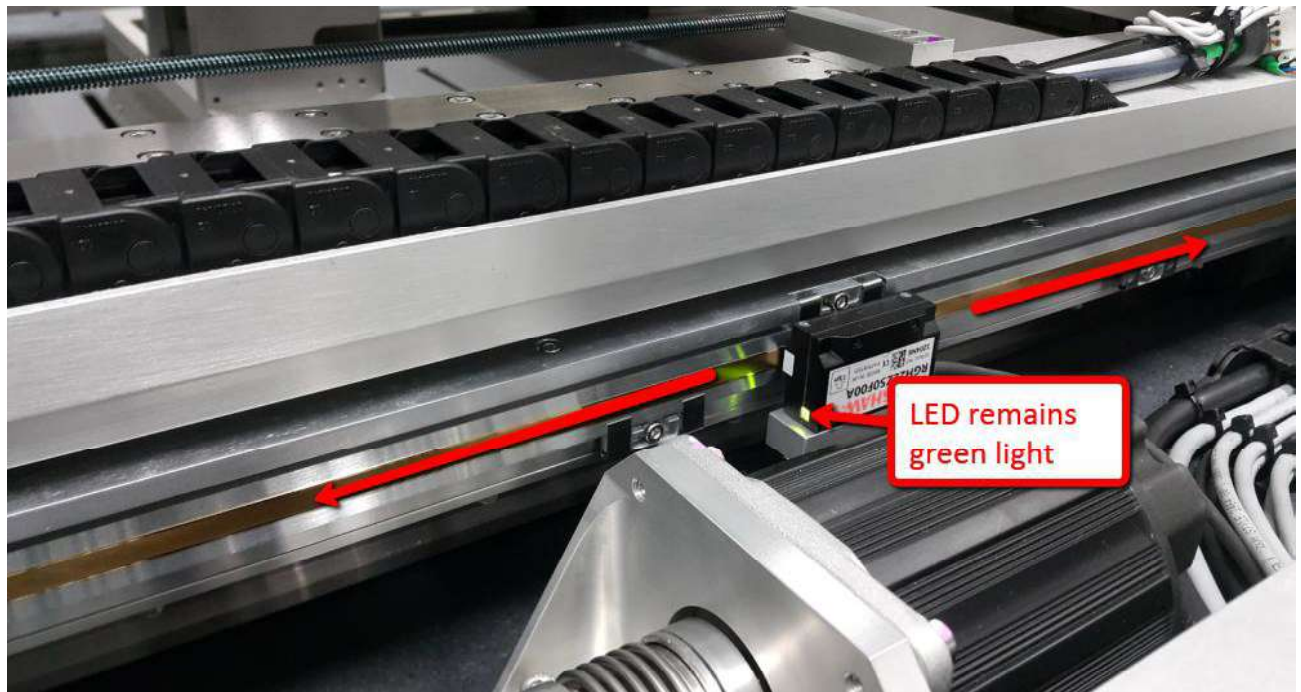


Figure 13 Y-Axis encoder checking

8. Close front access door.

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9. Open V810 > Services > Subsystem Status & Control > Panel Positioner > Initialize

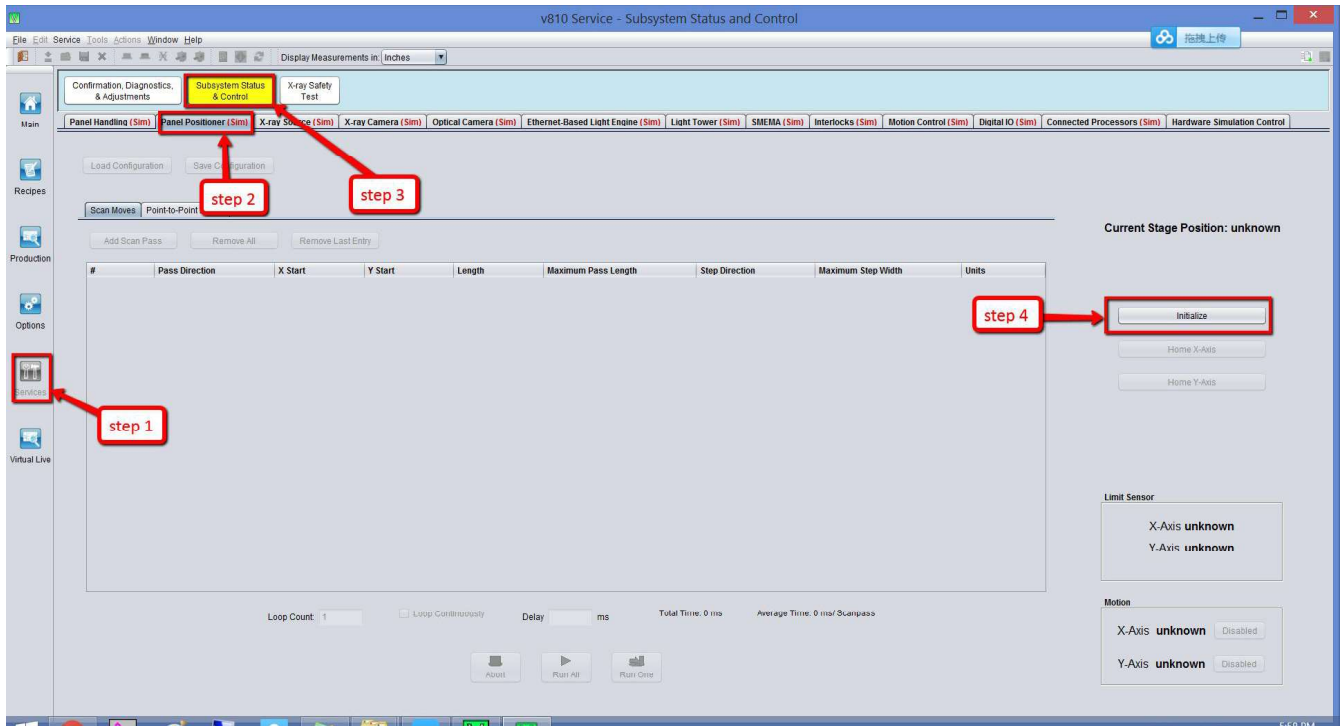


Figure 14 Initialize Panel Positioner

10. Open front access door.

11. Disable Y-Axis motion and pull Y-Axis stage & perform below checking.

- DO not move XY Stage and may check Limit sensor status at Panel Positioner where must stated as "Clear" during home position

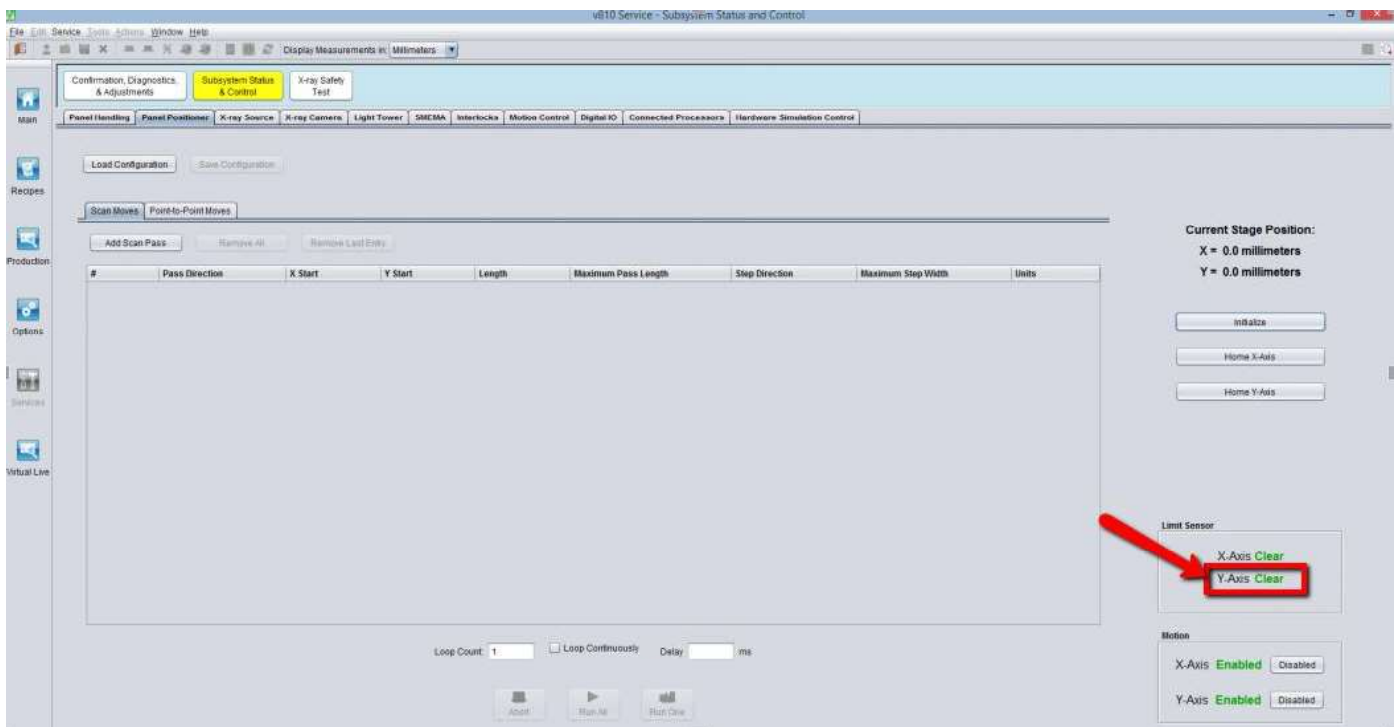


Figure 15 Y axis Limit Sensor (CLEAR)

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- b) Move Y axis stage towards minimum hard stop (Front side) & check Limit sensor where must stated as “ Blocked”.

Remarks: If result is not meet Step 11a ~ 11b requirement, may refer to “AXI V810 Y-Axis Limit Sensor Adjustment Procedure” for further adjustment.

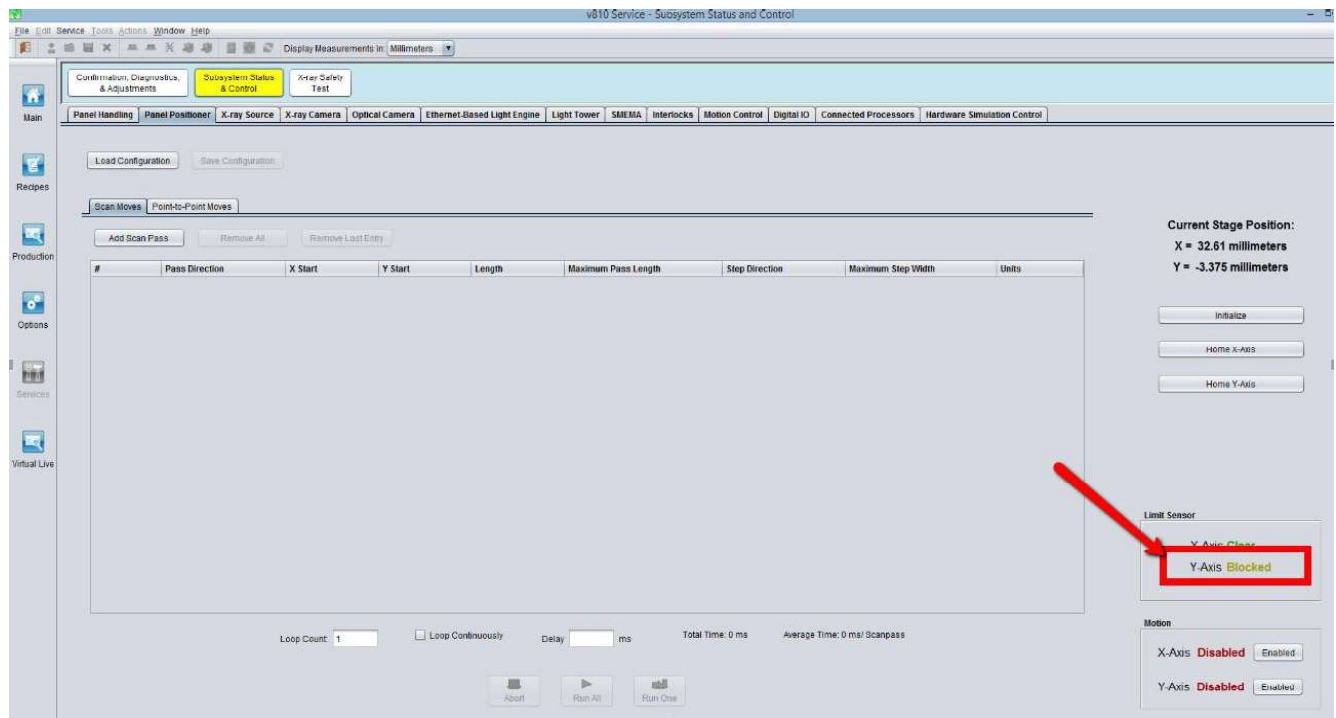


Figure 16 X axis Limit Sensor (BLOCKED)

- c) Move Y axis stage towards minimum hard stop(Front side).
d) Check the value for the Y-axis in current stage position must within 3.00mm ~ 3.30 mm.

Remark: If result is not meet to Step 11c ~ 11d requirement, may refer to “AXI V810 Y-Axis Home Sensor Adjustment Procedure” for further adjustment.

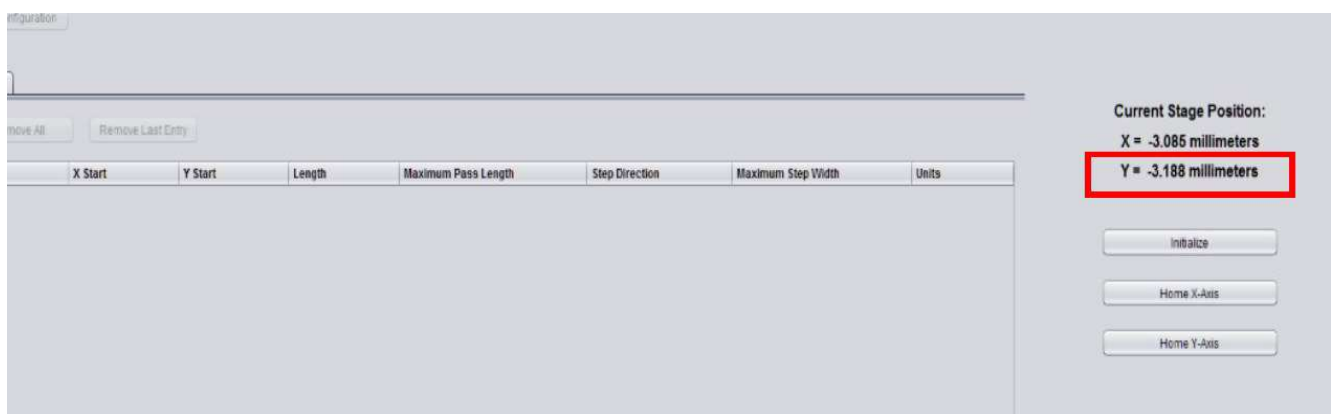


Figure 17 Y axis (Current Stage Position)

12. Run 3 loops of long scan and 1 loop of CD &A.